ELASTIC SECURITY

SIEM · EDR · Threat Hunting

TOOL INTRODUCTION: ELASTIC SECURITY

1. What Elastic Security Actually Is (Precise Definition)

Elastic Security is a **unified security analytics platform** that combines:

- **SIEM** (log-centric, environment-wide visibility)
- **EDR** (endpoint behavioral telemetry and response)
- **Threat hunting** (query-first, hypothesis-driven)

Critical distinction:

Elastic Security does **not** replace analyst reasoning.
It exposes **raw + enriched telemetry** and lets analysts decide.

This is why it fits your mindset.

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2. Where Elastic Security Sits in Security Architecture

Elastic Security is **not just an endpoint tool**.

It has **three visibility planes**:

1. **Endpoint plane** → Elastic Defend (EDR)
2. **Log plane** → SIEM (Windows, Linux, AD, network)
3. **Query plane** → KQL / EQL for investigation & hunting

Most EDRs hide (2) and (3). Elastic does not.

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3. Core Components (Vendor-Neutral Mapping)

3.1 Elastic Agent (Endpoint Sensor)

What it is

- Single agent installed on endpoints
- Replaces “beats” (Winlogbeat, Auditbeat, etc.)

- Runs in **user + kernel context**

What it does

- Collects endpoint telemetry:
    - Process execution
    - Command-line arguments
    - File operations
    - Registry activity
    - Network connections
- Enforces response actions **only when told**

What it does NOT do

- Decide intent
- Understand business context
- Automatically fix incidents

It is a **sensor + actuator**, not a brain.

3.2 Elastic Defend (EDR Capability)

This is the **EDR layer** inside Elastic Security.

Provides:

- Endpoint behavioral telemetry
- Detection rules (behavioral, not signature-only)
- Response actions:
    - Kill process
    - Isolate host
    - Quarantine (limited)
    - Prevent execution (policy-based)

Important:

Elastic Defend is **post-execution by design**, aligned with real EDR principles.

3.3 Elasticsearch (Data Store)

Role

- Stores **all telemetry**
- Time-series optimized
- Queryable at scale

This matters because:

- You can see **what alerts hide**
- You can investigate **without alerts**

3.4 Detection Engine

What it is

- Rule-based + behavioral logic
- Uses:
    - EQL (event correlation)
    - Threshold logic
    - Sequence-based detection

What it produces

- Detections (internal)
- Alerts (surfaced, compressed)

Remember:

Detection ≠ truth
Alert ≠ incident

3.5 Security UI (Interface, Not Intelligence)

Provides:

- Alerts
- Timelines
- Process trees
- Investigation views

Provides **zero judgment**.

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4. Elastic Security Vocabulary (Only What's Tool-Specific)

You already know generic EDR terms. These are **Elastic-specific mappings**:

Elastic Term	Actually Means
Event	Raw telemetry
Signal / Alert	Surfaced detection
Rule	Detection logic
Timeline	Analyst reconstruction view
Endpoint policy	Sensor behavior config
Isolation	Network cut at host level

If a term feels "high-level", always map it back to:
telemetry → **detection** → **alert** → **analyst decision**

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5. HOW ELASTIC SECURITY WORKS IN THE REAL WORLD (STEP-BY-STEP)

This is **exact**, not marketing.

Step 1 — Endpoint Activity Happens

Example:

- `winword.exe` spawns `powershell.exe`
- Encoded command executes
- Network connection is made

Step 2 — Elastic Agent Records Telemetry

- Process start event
- Command-line captured
- Parent–child relationship
- Network metadata

No judgment. No verdict.

Step 3 — Telemetry Is Indexed in Elasticsearch

- Events stored
- Correlated over time
- Available for queries

Step 4 — Detection Rules Evaluate Patterns

Example logic:

- Office → PowerShell
- Encoded command
- Outbound network

Rule evaluates **pattern**, not intent.

Step 5 — Alert May Be Created

Only if:

- Threshold crossed
- Confidence high enough
- Rule not suppressed

Many detections never surface.

Step 6 — Analyst Investigates

You:

- Open timeline
- Rebuild execution chain
- Validate user context
- Correlate with SIEM logs

Step 7 — Analyst Decides Response

Options:

- Do nothing
- Monitor
- Kill process
- Isolate host
- Escalate

Elastic **executes**, you **own the impact**.

Final Conclusion

Elastic Security is not an “EDR tool,” not a “SIEM dashboard,” and not an automated decision system. It is a **security analytics platform designed to expose reality**, not to hide it behind verdicts. Its core value is not prevention or alerting, but **visibility, correlation, and analyst control**.

Across this introduction, the same principle holds consistently:

- **Elastic Agent observes behavior**; it does not judge it.
- **Elasticsearch stores and correlates telemetry**; it does not decide intent.
- **Detection rules generate hypotheses**; they do not establish truth.
- **The Security UI visualizes data**; it does not think.
- **The analyst validates context, establishes truth, and owns response decisions**.

Elastic Security deliberately surfaces **raw and enriched telemetry** across endpoints, logs, and queries, giving analysts the ability to reconstruct events end-to-end—even when no alert fires. This is why it fits mature SOC workflows and why it is often misunderstood by those expecting automatic answers.

The non-negotiable takeaway is simple:

Elastic Security provides observations and hypotheses.
Analysts determine truth, impact, and response.

If this mental model is clear, Elastic Security becomes a powerful investigative platform where alerts are optional, hunting is natural, and response is deliberate. If this model is ignored, the platform will be misused as just another alerting tool—wasting its real strength.

With this foundation locked in, the next step is logical and safe: **building the lab**, where these principles will be validated through real telemetry, real behavior, and real analyst decisions.